AARE_BEC_WIDGETS · 2026-10-10 · ONE RUN, SAME MODEL, SAME SPEC

Smargon omega preset widget, built two ways

Two fresh agents built the same BEC plugin widget in separate worktrees. A followed the `bec-skills:bec-new-widget` skill. B ran your default workflow (ponytail hook + CLAUDE.md, no bec skills). Two blind reviewers then judged both branches. The deliverable is this comparison; both branches stay in place.

A · bec-new-widget skill · `smargon/a`

B · ponytail + CLAUDE.md · `smargon/b`

At a glance

Hand-written lines	Tests written	Build cost, tokens	Build time
756 **243**	**19** **4**	**217k** **119k**	**17.4 min** **5.4 min**
A is 3.1x B (widget + tests + conftest + yaml)	all green, random order, offscreen	A is 1.8x B	A is 3.2x B

Ponytail review: cuttable	BEC focused review
-354 **-10**	**approve, 4 fixes** **approve, 4 fixes**
lines the reviewer would delete	verdict per side

Numbers side by side

	A	B
Hand-written lines widget, tests, conftest, yaml	243	756
Whole diff vs main incl. regenerated client.py, Designer stubs	+374 / 11 files	+937 / 12 files
Tests added	4	19
Implementer tokens	119,126	216,925
Implementer wall-clock	5.4 min	17.4 min
Tool calls	23	31

Each row is its own scale: the larger of the pair spans the full track. Tokens, tool calls and durations are the harness usage numbers for each agent.

FILE	A LINES	B LINES	NOTE
`widgets/omega_control/omega_control.py`	339	106	A re-implements read target, Go/Stop; B con `PositionerBox` and preset grid.
tests (`test_omega_control.py`)	250	57	A: 19 tests in `tests/tests_bec_w` B: 4 tests in `tests/` .
tests conftest	148	65	Both had to vendor a r client: core 3.37.3 shi `bec_widgets.tests`
`device_configs/smargon_omega.yaml`	19	15	Same SimPositioner, r (continuous axis). A a `precision: 2` and a tag.
generated by `bw-generate-cli`	182	132	client.py, designer_plu Designer plugin + regi pyproject. Larger in A USER_ACCESS is large
dev loop	2	2	A: one line each in de `Dockerfile.rhel9` . only, container entryp without the axis.

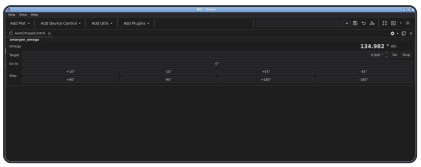
AGENT	TOKENS	TOOL CALLS	MINUTES	OUTPUT
A implementer (bec-new-widget)	216,925	31	17.4	widget, 19 tests, yaml, both dev loops
B implementer (ponytail)	119,126	23	5.4	widget, 4 tests, yaml, dev.sh
Reviewer 1 (ponytail-review, both)	107,538	21	6.4	28 cut lines, 2 net totals, pick
Reviewer 2 (bec-focused-review, both)	204,409	72	27.2	8 verified findings with probes, cleanup lists, pick

Harness usage per agent. Reviewer 2 fans out its own finder sub-agents and runs probes, which is where its cost goes.

Running in bec-app on the RHEL9 container

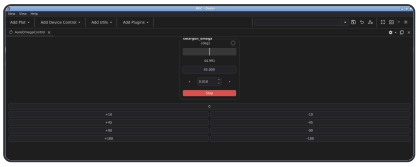
Each branch launched `bec-app` from its worktree, the widget was added over RPC with `Docks.new("AareOmegaControl")`, then one relative +45 move was sent through the widget's own RPC method. Both independently named the class `AareOmegaControl`.

A readback, target + Go/Stop, "Go to" row, "Step" grid



134.982° idle after `move_relative(45)`. Status flag from `smargon_omega_motor_is_moving`, moves via `submit_task(scans.mv)`.

B core PositionerBox + preset grid



44.991 → 45.000 mid-move after `move_by(45)`. Readback, setpoint, tweak and Stop come from `PositionerBox`; the unit label is overridden to [deg].

Reviewer 1: ponytail-review (over-engineering only)

Blind to which workflow built which branch. One line per finding: location, what to cut, what replaces it. Tags: delete yagni shrink and native (use the platform). Cost of this review: 108k tokens, 21 tool calls, 6.4 min for both branches.

A 26 findings · net -354 lines possible

```
omega_control.py:L117-139,L210-220: native: readback label + moving flag + target spinbox + Go/Stop re-implement bec_widgets PositionerBox (readback, setpoint, stop, units, moving). Compose `PositionerBox(parent=self, client=self.client, device=device)`, keep only the preset grid. -40.
```

```
omega_control.py:L166-206: yagni: `device` SafeProperty with re-subscribe + `_init_from_device` park/revive + `_set_controls_enabled`. Spec fixes `smargon_omega`; PositionerBox(device=...) owns
```

B 2 findings · net -10 lines possible

```
omega_control.py:L90-92: delete: `cleanup()` override calling `self.box.cleanup()`. `BECWidget.cleanup()` already close()s every BECWidget child (bec_widget.py:412-418). -3.
```

```
omega_control.py:L65,L72-78: shrink: 8-line row/col walker so "0" gets a full row and +x/-x pair up. `self._preset_layout.addWidget(btn, i // 2, i % 2)` with `enumerate`; "0" then shares its row with "+10". -7.
```

the subscription, a missing device surfaces as the

Reviewer 1 comparison

- B leaner: 228 hand-written lines vs 737. B composes core PositionerBox for readback, setpoint, stop, units, moving; A re-implements all of it plus device re-targeting, park/revive, threaded moves and theme colours.
- A has, B lacks: the Dockerfile.rhel9 `add_to_session` line, a Qt Designer editable preset property, a top-level-widget leak check in conftest, tooltips per preset, `precision: 2` in the yaml.
- B has, A lacks: typed setpoint, tweak and stop for free via PositionerBox, the [deg] unit label, a flat constructor.
- Presets: A = two float lists inside a pydantic ConnectionConfig, set via ctor kwargs, config dict or `SafeProperty(str)` strings. B = one `(label, degrees, relative)` list, set via the module constant, ctor `presets=` or `set_presets()` over RPC.
- Tests: A's 19 mostly guard A's own extras (codec, three config paths, device swap, park/revive, theme, core cleanup). B's 4 cover the spec at a fifth of the size.
- **Pick: B.** Port from A: the Dockerfile.rhel9 yaml line, and A's top-level-widget leak fixture (about 12 lines) into `tests/conftest.py` . Then apply B's own two cuts.

Reviewer 2: bec-focused-review (correctness, Qt/thread/dispatcher pitfalls, tests)

Blind as well. Judges only what the change introduces; every finding is tagged `introduced` or `exposed` , pre-existing repo problems go to a separate list. Both branches got **approve with fixes**. Author claims were checked by regenerating the Designer files (byte-identical on both) and by lifecycle probes. Cost of this review: 204k tokens, 72 tool calls, 27.2 min for both branches (vs 108k, 21, 6.4 min for reviewer 1).

A approve with fixes · 4 introduced (3 confirmed, 1 plausible)

1. [introduced, CONFIRMED] omega_control.py:321 apply_theme: Go/Stop accent colours never applied at construction. BECWidget only connects apply_theme to theme_changed; AareGridScanControl calls self.apply_theme(get_theme_name()) at the end of __init__. Probe: GO STYLESHEET AT INIT: ''.

2. [introduced, PLAUSIBLE]

B approve with fixes · 4 introduced (3 confirmed, 1 plausible)

1. [introduced, CONFIRMED] Dockerfile.rhel9:67 untouched: smargon_omega.yaml is loaded by dev.sh only. In the container the session has no smargon_omega; the widget shows the PositionerBox placeholder, every preset click raises KeyError -> popup. (The screenshot above loaded the yaml by hand.)

2. [introduced, CONFIRMED] omega_control.py:60-77

```
omega_control.py:238 _move:
failure path covers client-
side exceptions only. A scan-
server rejection (limit,
disabled device) comes back
async as
RequestResponseMessage(accepte
d=False); on_failed never
```

Checked and fine: one bound-method dispatcher slot released by BECWidget.cleanup() and swapped on device change; worker signals queued to the GUI thread; MRO, SafeSlot on every slot, USER_ACCESS extends the base list; plugin discovery and Designer registry resolve; yaml keys valid, (0,0) limits = unlimited; every preset path tested; leak + thread + popup guards active in the fallback conftest; random order stable over 4 runs.

```
set_presets deletes the old
buttons before validating the
new list; a malformed entry
leaves partial state. Probe:
set_presets([('+30',30,True),
('bad','x',True),
('0',0,False)]) -> buttons
['+30']. Fix: normalise first,
```

Checked and fine: widget owns no subscriptions, timers or threads (readback belongs to the child PositionerBox, released on close); scans.mv synchronous inside SafeSlot exactly like core; MRO and slots correct; hide_device_selection is a real 3.37.3 property; plugin discovery and generated files identical to bw-generate-cli output; yaml valid; DEFAULT_PRESETS exactly the spec list; random order stable.

Reviewer 2 comparison

- No happy-path correctness bug in either. A's findings: cosmetic theme-at-init, an over-claimed failure path, silent preset typos, formatting. B's: the container config gap, partial state on bad presets, non-persisted presets, formatting.
- Lifecycle: A owns one dispatcher slot and runs moves on the QThreadPool, correct but heavier than needed. B delegates everything to the embedded PositionerBox, zero own subscriptions or threads, one redundant double cleanup. Both correct.
- Tests: A 19 (readback, every preset path, config paths, device swap, unknown device, failure path, close idempotence) plus core-equivalent leak/thread/popup guards. B 4, no lifecycle test, no guards.
- Spec fit: A meets every bullet. B misses the Dockerfile and half of "easily configurable" (no persistence).
- **Pick: A.** Port from B: the +x/-x pairwise grid; embedding PositionerBox instead of the hand-rolled readback/Go/Stop row; the synchronous scans.mv call (and drop the thread-pool warm-up hack with it).

Pre-existing, not blamed on either branch

- Core `PositionerBoxBase._init_device` passes an unguarded `read(cached=True)` None into the readback slot.
- `AareGridScanControl` and `AareRasterPanel` already replace `USER_ACCESS` instead of extending it.
- The generator's `client.py` import grouping fails the repo's isort config on both branches and on main.

What each workflow did and where it rubbed

A bec-new-widget skill

- **Shape.** `BECWidget + QWidget`, `PLUGIN`, `ICON_NAME`, 12-entry `USER_ACCESS`, `@SafeSlot` everywhere, `SafeProperty(str)` for Designer-editable presets, pydantic `AareOmegaControlConfig`, moves through `submit_task(..., on_failed=)`. Device re-targetable at runtime; unknown device parks the controls.
- **Friction.** Skill assumes core ≥ 3.38 (`bec_widgets.tests.fixtures`, `_call_later`); venv has 3.37.3, so a 148-line fallback conftest with a TODO. `bec-plugin-manager create widget` needs a clean tree + prompts, so the layout was copied by hand. Fake client is a singleton, so a per-test `mv` patch fixture.
- **Verification.** 22 tests, 3x random order, black/isort clean, plus an offscreen render with the fake client. Added the yaml to both dev loops.
- **Skipped.** No live run (forbidden), Designer listing, safety-audit probes, the stale `aare_demo_widget` stub.

B ponytail + CLAUDE.md

- **Shape.** Composes core `PositionerBox(device="smargon_omega"` and adds a preset grid. Presets = one module constant of `(label, deg, relative)`, ctor kwarg, `set_presets()` RPC. 5-entry `USER_ACCESS`. Moves call `scans.mv` inline, same as PositionerBox. One `# ponytail:` comment marks the [deg] unit override.
- **Friction.** Same fixtures gap, answered with a 65-line conftest (no leak checks, said so in the docstring). Fake `DEVICES` lacks smargon_omega, so a `FakePositioner` is added. SimPositioner reports egu "mm". `bw-generate-cli` output needed isort.
- **Verification.** 7 tests, black/isort clean on new files. dev.sh loads the yaml; Dockerfile.rhel9 not touched (not asked).
- **Skipped.** No Designer property for presets, no leak checks, no live run.

Verdict

What the data says

- Same model, same spec, same env facts. B shipped a working widget in a third of the time and lines, with the reviewer that hunts bloat finding almost nothing to cut.
- A shipped a more defensive widget: runtime device swap, failure reporting in the status label, Designer-editable presets, leak-checking conftest. The ponytail reviewer counts most of it as unrequested.

Suggested merge

- Take `smargon/b` as the base: same working widget, a third of the code, built in a third of the time.
- Port from A: the `Dockerfile.rhel9` yaml line (both reviewers flagged it); the top-level-widget leak fixture; `precision: 2` in the yaml.
- Apply B's two ponytail cuts (drop the `cleanup()` override, flatten the row walker) if you agree with them.
- Then take reviewer 2's fixes for B:

- Both hit the same wall: the skill's test fixtures target a newer bec_widgets than the venv has. The skill-driven agent spent more tokens working around it than the plain agent.
- The reviewers split exactly along their philosophies: the bloat hunter picks B, the BEC contract reviewer picks A. Reviewer 2 found no happy-path bug in either; its A-vs-B gap is persistence of presets, the container yaml line and test guards.
- Reviewer 2's own port list (embed PositionerBox, synchronous mv, pairwise grid) walks A toward B's shape. The two verdicts converge on: B's structure, plus A's config persistence and input validation.

Then take reviewer 2's fixes for B: validate before rebuilding in `set_presets` (and `popup_error=True`), store presets in a `ConnectionConfig` field with a `SafeProperty(str)` if saved layouts or Designer must carry them, fix the black line, extend `USER_ACCESS` instead of replacing it.

- Add `QMessageBox` suppression and the leak check to `tests/conftest.py` (A's fallback shows the shape).

Method and caveats

▼ Spec handed to both implementers (identical text)

```
Build a BEC plugin widget in this repo (aare_bec_widgets) for the smargon goni
- Shows current omega angle of device `smargon_omega`, moves it.
- Preset moves: 0 deg (absolute), and relative +10, -10, +45, -45, +90, -90, +1
- Presets must be easily configurable.
- Mock motion: `smargon_omega` is an ophyd_devices.SimPositioner. Add its yaml
  aare_bec_widgets/device_configs/ (see demo_sample_camera.yaml for the patter
- Widget must be discoverable by bec-app as a plugin widget (entry point bec.wi
  -> aare_bec_widgets.bec_widgets.widgets already exists).
- Leave runnable tests. Run them headless. Commit on your branch when green.
Env: shared venv ../bec_widgets/.venv, PYTHONPATH=<worktree> on every call, no
no files outside the worktree, no skill or workflow names in code or commits.
```

▶ Protocol

▶ Caveats

Decision by the orchestrating agent

> **Build widgets in this repo with your workflow (ponytail + CLAUDE.md). Gate**

each merge with bec-focused-review. Reach for bec-new-widget only for